

MSI411-MATHEMATICAL STATISTICS

WEEK: 1 (PROBABILITY FOUNDATIONS)

INSTRUCTOR: CARLES QUILIS ALFONSO, CQU@BTH.SE

REFERENCE TEXT: WALPOLE, MYERS, MYERS, AND YE (9TH ED.), *PROBABILITY & STATISTICS FOR ENGINEERS & SCIENTISTS*

Outline:

- About the course logistics
- Introduction to probability & counting (permutations and combinations)
- Fundamental probability laws
- Conditional probability
- Suggested reading: Walpole et al., Chapters 1 (optional) and 2.1–2.6



ABOUT THIS COURSE

- **Exam dates:** June 07
- **Registration period:** 08 May 2025-24 May 2025
- **Reference textbook:** *Walpole et al., Probability & Statistics for Engineers & Scientists*, 9th ed., Pearson
- **Topics:** Probability fundamentals, sampling distributions, estimation, hypothesis testing, regression, experimental design, etc.
- **Assessment:** Written exam, exercises, etc.
- **Tip:** Keep up with reading in Walpole; do the practice exercises!



MOTIVATION—WHY PROBABILITY?

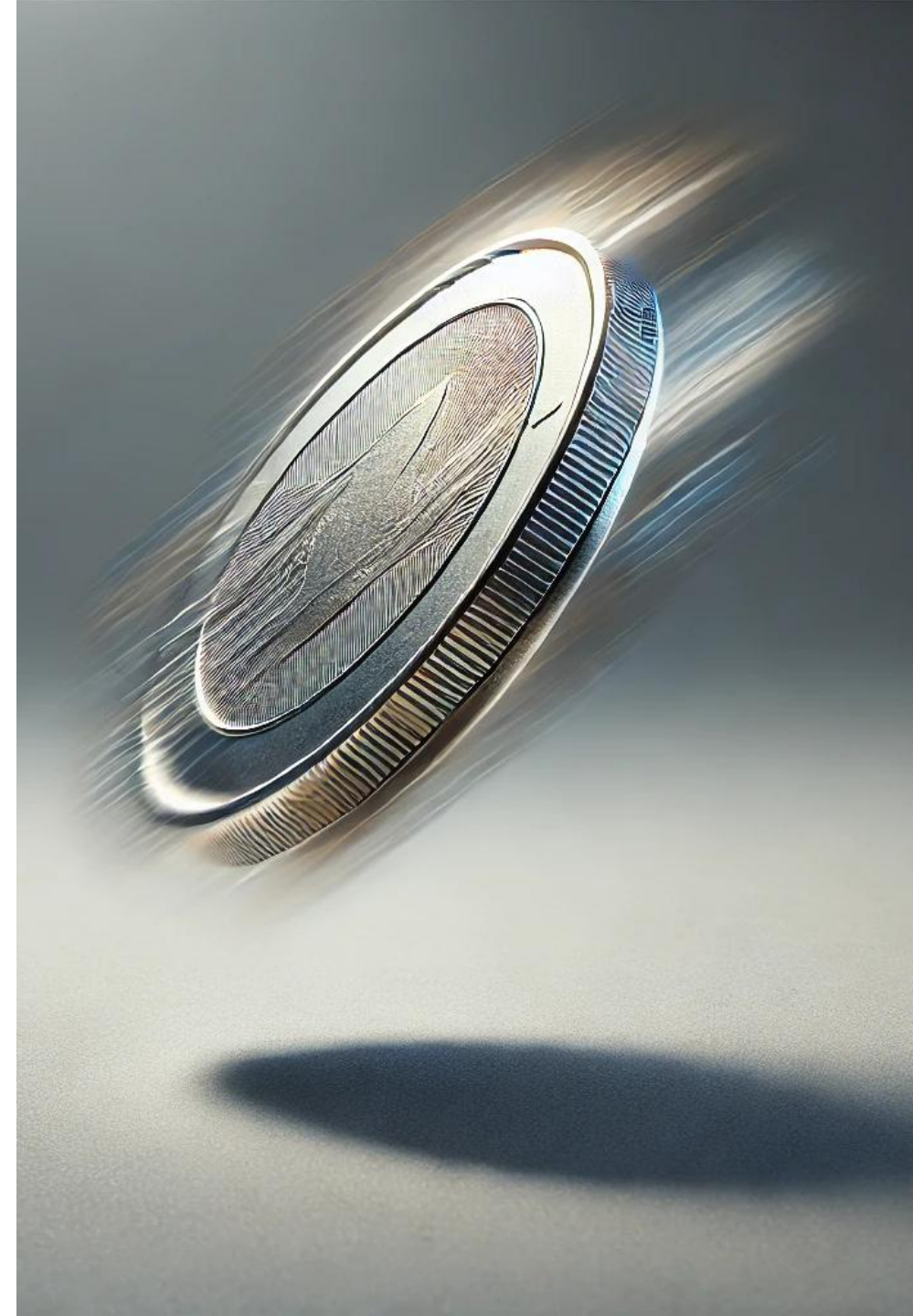
- **Nature of “randomness”:** Real processes often have variability (e.g., manufacturing defects, measurement error, daily weather).
- **Engineers and scientists:**
 - Need to quantify uncertainty.
 - Make decisions with incomplete information.
- **Central question:** “How do I measure likelihood?” and “What if my observations deviate from the expected pattern?”
- **Mini Example:** The yield in a chemical process can vary from batch to batch. Probability helps characterize that variation, enabling you to make predictions and optimize processes.

DEFINITIONS

- Experiment: A process that generates a set of possible data outcomes.
 - Example: Rolling a die, measuring battery lifetimes, surveying opinions.
- Sample space (S): The set of all possible outcomes of the experiment.
 - Example (dice): $S = \{1, 2, 3, 4, 5, 6\}$.
- Sample point: An individual outcome in S .
- Event: Any subset of the sample space. Often denoted $A \subset S$.
 - Example: $A = \{\text{even numbers}\} = \{2, 4, 6\}$.

SAMPLE SPACE EXAMPLES

- Tossing a coin twice: $S=\{HH,HT,TH,TT\}$.
- Engineering:
 - Inspect an item: $S=\{\text{defective (D), nondefective (N)}\}$.
 - Sample 3 items: $S=\{DDD, DDN, DND, \dots, NNN\}$.
- Class Exercise: List the sample space for “roll two dice and observe sum.”
- (There are 36 equally likely sample points—but 11 distinct possible sums.)



COUNTING TECHNIQUES—PERMUTATION & COMBINATION

I. Fundamental Counting Principle:

- If an experiment or procedure can be broken into sequential steps (or stages), and
 - Step 1 can be done in n_1 ways,
 - Step 2 can be done in n_2 ways,
 - ... and so on,
- then the total number of ways to complete all steps is: $n_1 \times n_2 \times \dots$.

Example: If in an experiment you first select 2 out of 5 possible chemical additives ($\binom{5}{2}$ ways), then choose which of 3 possible temperatures to run (3 ways), total possible test setups = $\binom{5}{2} \times 3 = 10 \times 3 = 30$.

COUNTING TECHNIQUES—PERMUTATION & COMBINATION

- Permutations: When Order Matters

- The number of ways to *arrange* or *order* “ r ” objects out of “ n ” distinct objects.
- Denoted by nPr or $P(n,r)$
- Formula: $nPr = \frac{n!}{(n-r)!}$

Example: Suppose 3 awards (research, teaching, service) are given to a class of 25 grad students, each student can receive at most one. (The order matters because is not the same result that you receive one award or another one) Then:

$${}_{25}P_3 = \frac{25!}{(25-3)!} = \frac{25 * 24 * 23 * 22 * \dots}{22 * 21 * 20 * \dots} = 25 * 24 * 23 = 13,800$$

COUNTING TECHNIQUES—PERMUTATION & COMBINATION

- Combinations: When Order Does *Not* Matter

- The number of ways to *choose* “ r ” objects out of “ n ” distinct objects *without* regard of the order.

- Denoted by nCr or $\binom{n}{r}$

- Formula: $nCr = \frac{n!}{r! (n-r)!}$

Example: If 3 identical awards are given and it doesn’t matter who’s first, second, or third, then the count of ways to pick 3 winners out of 25 is $\binom{25}{3} = 2300$

Tip: Look for wording clues—“arrange,” “rank,” or “order” implies permutations. “Select,” “choose,” or “pick” (with no mention of arrangement) implies combinations.

PROBABILITY AXIOMS & BASIC DEFINITIONS

■ Three Axioms of Probability

1. Axiom 1: $P(S)=1$

1. The probability of the entire sample space S is 1.

2. Axiom 2: For any event $A \subseteq S$, $P(A) \geq 0$.

1. No negative probabilities.

3. Axiom 3 (Additivity): If two events A and B are *disjoint* or *mutually exclusive* ($A \cap B = \emptyset$), $P(A \cup B) = P(A) + P(B)$.

1. They cannot occur simultaneously, so the probability of their union is the sum.

PROBABILITY AXIOMS & BASIC DEFINITIONS

■ Immediate Consequences

- **Empty Set:** $P(\emptyset) = 0$.
 - The “impossible event” has probability zero.
- **Complements:** $P(A^c) = 1 - P(A)$.
 - The probability that A does *not* happen is 1 minus the probability it does. **Also denoted by A'**
- **Additive Rule for Overlapping Events:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
 - Subtract the overlap once to avoid double-counting it.

PROBABILITY AXIOMS & BASIC DEFINITIONS

■ **Example** (Die Roll):

- Let $S=\{1,2,3,4,5,6\}$ for a fair 6-sided die.
- The event $A=\{\text{"roll an even number"}\}$ has 3 outcomes ($\{2,4,6\}$), so $P(A)=3/6=1/2$.
- The complement A^c or A' is "roll an odd number," so $P(A')=1-P(A)=1/2$.

■ **Mutually Exclusive vs. Independent (Preview)**

- *Mutually Exclusive* (disjoint): The events cannot happen at the same time ($A \cap B = \emptyset$).
- *Independence* (discussed in a later slide) concerns whether knowing one event occurs changes the probability of the other.

CONDITIONAL PROBABILITY

- **Definition:** If A and B are events with $P(A) > 0$, the *conditional probability* of B given A is
 - $P(B | A) = \frac{P(A \cap B)}{P(A)}$
 - This “restricts” our perspective to the sample space where A has already happened and measures the fraction of those outcomes that also satisfy B .
- **Interpretation:**
 - “If we already know event A occurred, how does that influence the probability of B ?”
 - Real-World Example: If 10% of all factory items are defective, but we learn that an item came from a *specific* production line or shift, the probability might change. We recalculate using data restricted to that subset.
- **Link to the Intersection:**
 - We can rewrite $P(A \cap B) = P(A) P(B|A)$.
 - In words: The probability of A and B both happening is the probability of A , times the conditional probability that B also happens once A is known.

CONDITIONAL PROBABILITY

■ Illustrative Example (Cards):

- Sample space S : a deck of 52 cards.
- Let A = “drawn card is a spade” and B = “drawn card is the ace of spades.”
- $P(A) = 13/52 = 1/4$.
- $P(A \cap B) = 1/52$ (the single ace of spades).
- Then: $P(B | A) = \frac{1/52}{1/4} = \frac{1}{13}$,
- because once you *know* it's a spade (13 possibilities), only 1 of those is the ace.

■ Key Takeaways

- The **probability axioms** form the foundation for all probability computations.
- **Complement rules** and **additivity** help handle events systematically.
- **Conditional probability** quantifies how knowledge of one event affects the probability of another.
- Real-life contexts: reliability engineering, medical tests, quality control, etc., rely heavily on conditional probability ideas.



INDEPENDENT EVENTS

- **Definition:** Events A and B are *independent* if $P(A \cap B) = P(A)P(B)$.
- **Equivalent:** $P(B|A) = P(B)$ or $P(A|B) = P(A)$.
- **Example:** Rolling a fair die and flipping a coin—results do not affect each other.
- **Caution:** Independence is *not* the same as mutual exclusivity. Mutually exclusive events *cannot* occur together, whereas independent events *do not influence* each other's likelihoods.

CLASS ACTIVITY—ROLL THE DICE

1. Roll two dice simultaneously (or use an online dice simulator).
2. Let event A = “first die is a 3” and event B = “sum is 7.”
3. Are A and B independent?
 1. Hint: Check if $P(A \cap B) = P(A)P(B)$.

Solution:

$$P(A) = 1/6.$$

$$P(B) = 6/36 = 1/6.$$

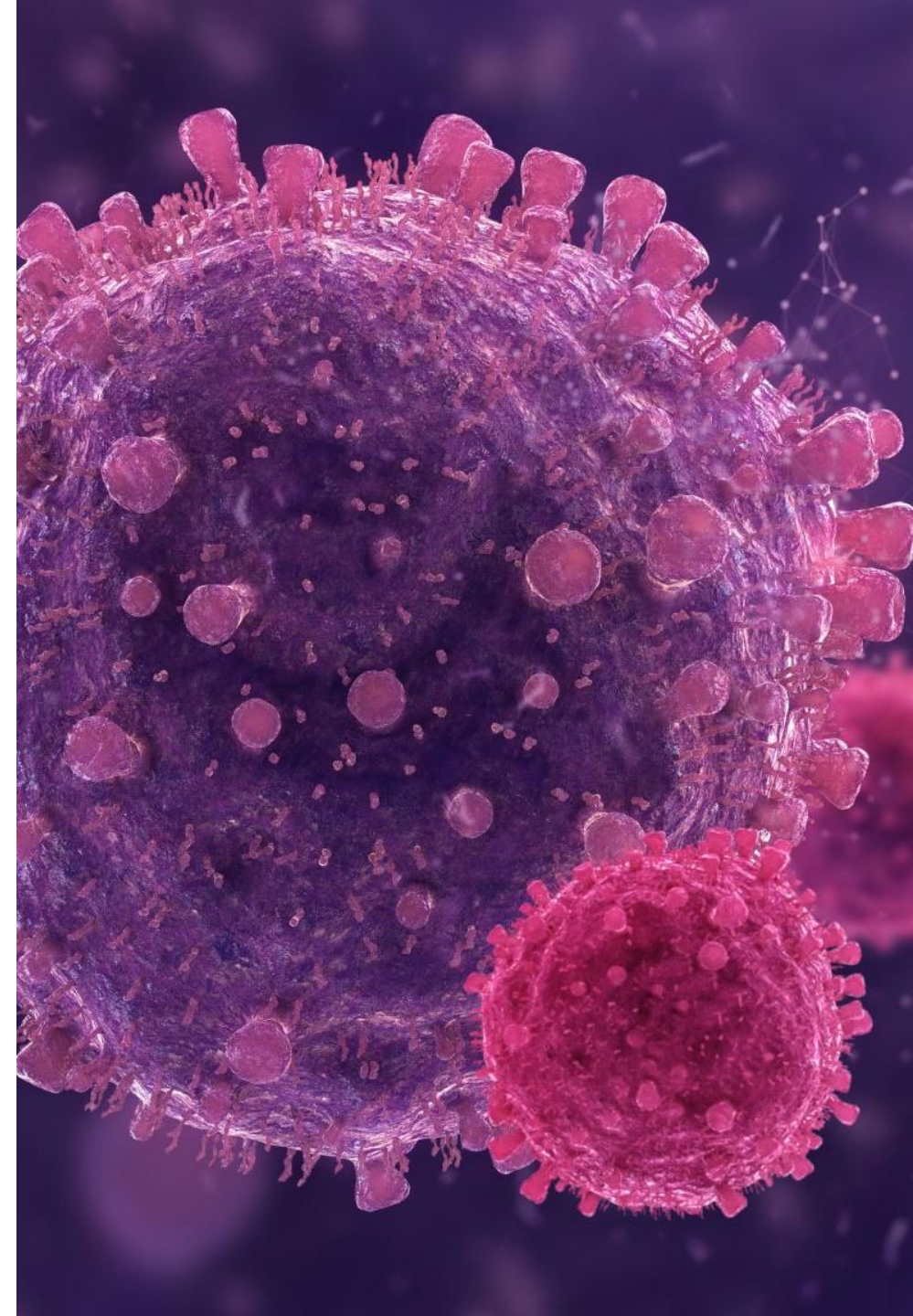
$P(A \cap B)$? That’s “first die is 3, sum is 7,” i.e. second die must be 4. Probability = $1/36$.

So $P(A \cap B) = 1/36$, but $P(A)P(B) = (1/6)(1/6) = 1/36$.

So yes, they are independent in this example.

EXERCISE

- Researches indicate that approximately 49% of all infections involve anaerobic bacteria. Additionally, 70% of all anaerobic infections are polymicrobial, i.e. involve more than one anaerobe. What is the probability one infection involves both anaerobic bacteria and is polymicrobial?
- Solution:
 - Infections involving anaerobic bacteria: $P[A] = 0.49$
 - Anaerobic infection is polymicrobial: $P[B|A] = 0.70$
 - Anaerobic and polymicrobial: $P[A \cap B]$?
 - $P[A \cap B] = P[B|A] \cdot P[A] = 0.70 \cdot 0.49 = 0.343$



SUMMARY, PRACTICAL NOTES & COMMON MISTAKES

1. Sample space & events: Provide the foundation for all probability analyses.
 2. Counting methods: Permutations & combinations help determine event sizes in finite cases.
 3. Probability axioms: Let you systematically compute event likelihoods.
 4. Conditional probability & independence: Key for multi-step processes and reliability analysis.
- Common mistakes:
 - **Forgetting the condition:** “Conditional probability” is easy to overlook, leading to erroneous results.
 - **Assuming independence:** Not all seemingly separate events are truly independent (check real-world processes carefully).
 - **Mixing up mutually exclusive & independent:** E.g., if events can’t happen together, that does not necessarily imply independence.